

Ananya Gulati

PRINCIPAL SOFTWARE ENGINEER

Navi Mumbai, India | +91 8574564055 | ananya97.mail@gmail.com | ananyagulati.me | github.com/ananya-gulati |
linkedin.com/in/ananya-gulati

PROFESSIONAL SUMMARY

Results-driven **Principal Software Engineer** with 6+ years of expertise designing and optimizing high-performance backend systems. **AI-First software engineer** actively orchestrating autonomous developer agents and LLM scripts to accelerate feature design, logging audits, codebase migrations, and system diagnostics. Proven record of scaling ingestion architectures and collaborating with global business leads to deliver technical schemas for highly configurable enterprise modules.

PROFESSIONAL EXPERIENCE

GEP Worldwide — Principal Software Engineer January 2026 – Present

- Lead technical schema creation and overall execution design for complex, configurable enterprise application modules, improving framework reusability and platform capabilities.
- Integrate agentic AI developer workflows into daily sprint workflows, boosting debugging iterations and custom scripting velocity by **35%**.

GEP Worldwide — Senior Software Engineer July 2020 – December 2025

- Architected and scaled **5+ core microservices** handling client transactions, driving end-to-end development, API definitions, and legacy system refactoring.
- Improved inbound request processing speed by **10x** through staging logic optimizations and adoption of bulk processing design patterns.
- Developed an Azure Databricks native ETL pipeline processing **150M+ raw time-series data points** from CSV files into MongoDB, incorporating custom validation and processing to support downstream analytics.
- Orchestrated a large-scale data migration of **3M+ records** in MongoDB using Python on Databricks, reducing data onboarding latency while maintaining **100% data integrity**.
- Engineered dynamic database-driven HTTP execution services, reducing development turnaround time by **30%** and boosting core application configurability by **20%**.
- Designed modular object validation engines, cross-app notification frameworks, and reusable workflow plugins, saving developers **15+ hours weekly** in boilerplate code setup.
- Coordinated cross-functional teams to diagnose and resolve **30+ critical and blocker production support incidents**, systematically optimizing performance bottlenecks with minimal business disruption.

Illinois Institute of Technology — Research Intern (Chicago, USA) May 2019 – July 2019

- Cleaned and mapped gigabytes of tabular genetic dataset using Python (Pandas & NumPy) to identify promising chemical combinations for Alzheimer's therapeutics.
- Designed statistical regression models to evaluate chemical compound efficacy, significantly reducing chemical compound shortlisting time.

TECHNICAL SKILLS

Languages: C# / .NET, C++, Java, Python, TypeScript, JavaScript, SQL, HTML5 / CSS3

Backend & Systems: ASP.NET WebAPI, Node.js, Express, Microservices, RESTful APIs, OOP, Camunda

Databases & Tools: MySQL, MongoDB, Elasticsearch, Git, Docker, Kubernetes, AWS, Databricks

Methodologies: Architecture Design, Performance Optimization, Agile/Scrum, Code Review, Mentorship

EDUCATION

Birla Institute of Technology, Mesra — Bachelor of Engineering in Information Technology 2016 – 2020

CGPA: **7.78 / 10.00**

SELECTED PROJECTS

Time Series Forecasting Application .NET Core, MongoDB, SQL

- Developed and scaled a super-configurable forecasting engine for buyer-supplier stock movements, optimizing read/write speeds for large time-series datasets.

Quality Management Platform Node.js, .NET Core, Angular, MongoDB

- Designed an enterprise low-code evaluation system enabling modular supplier-level quality evaluations, reducing buyer review overhead.

HONORS & AWARDS

National Talent Search Scholar (NCERT): Prestigious scholarship awarded by the Government of India (2014).

Smart India Hackathon Finalist: Onsite national-level team finalist in software edition (2019).

GEP SPOT Recognition: Received for demonstrating project ownership and platform delivery (2021).

GEP Change Agent: Awarded for driving architecture optimizations and platform enhancements (2020).